

Project Initialization and Planning Phase

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template)

A well-articulated customer problem statement allows the team to find the ideal solution for the target user's challenges. Below, the "customer" is the student using self-paced or independent study material outside a live classroom. Five problem statements are captured, one per emotional state the project's classifier is designed to detect (Frustrated, Confused, Curious, Confident, Bored) — this is the empirical basis for why the solution needed emotion detection rather than a single generic help flow.

PS	I (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A student attempting the same homework problem repeatedly	solve it using the method taught in class	the same approach keeps failing and they don't know why	there's no one available at that moment to explain the gap a different way	frustrated, and close to giving up
PS-2	A student encountering an unfamiliar concept for the first time	understand what the concept even means before attempting any problems	the explanation assumes background knowledge they don't have yet	there's no adaptive, one-on-one explanation available outside class hours	confused, and unsure where to even start
PS-3	A student who has just finished the assigned material	explore the topic further and ask "what if" questions beyond the syllabus	there's no easy way to get a quick, tailored answer without derailing into unrelated searching	there's no low-friction way to extend learning right when curiosity is highest	curious, but sometimes unsupported in following it
PS-4	A student who has just solved several practice problems correctly	get quick confirmation that their understanding is solid before moving on	they only find out if they were truly correct much later, via a graded assignment	there's no immediate, light-touch validation loop	confident, but that confidence isn't reinforced in the moment
PS-5	A student working through material that reads as repetitive or already-familiar	stay engaged long enough to actually finish the session	the same static content and pace is used regardless of how they're responding to it	there's no way for study material or support to adapt to how a student is feeling	bored, and increasingly disengaged from the material